

P001 - Order Queue Screen Specification

Module: PSP Operations Portal **Screen ID:** P001 **Route:** /psp/orders **Version:** 1.0 **Last Updated:** 2026-01-01
IEEE 830 Reference: Section 3.2 - Functional Requirements **SUPP Reference:** SUPP-016 (Fulfillment - Order Processing)

1. Screen Overview

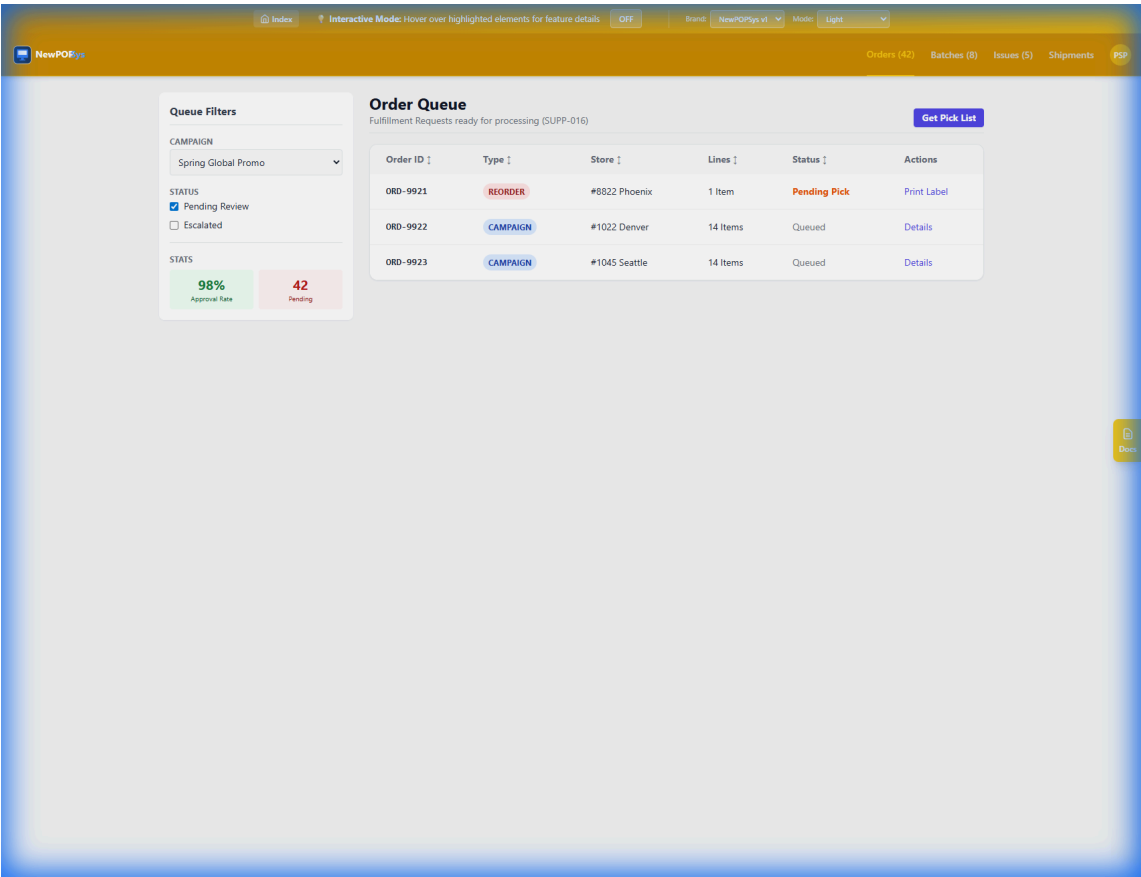
1.1 Purpose

The Order Queue screen provides PSP Production Operators with a centralized view of all store orders requiring fulfillment action. It enables order acknowledgment, status tracking, and drill-down access to order details for production processing.

1.2 Business Context

This screen serves as the primary work queue for PSP fulfillment operations. Orders flow from campaign assignments through to shipment, with this screen managing the critical handoff between order generation and production.

1.3 Screenshot Reference



1.4 Navigation Path

PSP Portal → Orders (sidebar) → /psp/orders

1.5 Related Screens

Screen	Relationship
P002 Shipments	Orders link to shipments after production
P003 Issues	Issues may trigger reorders appearing in queue

2. User Roles & Permissions

2.1 Authorized Roles

Role	Access Level	Permissions
PLATFORM_ADMIN	Full	View all, update status, bulk actions
PSP_ADMIN	Full	View all, update status, bulk actions
PSP_OPS	Operational	View all, update status, bulk actions
Support Agent (PSP_OPS + support_scope)	Read-Only	View only, no status updates

2.2 Permission Requirements

Requirement ID	Description	Roles
REQ-P001-SEC-001	User SHALL be authenticated with valid JWT containing tenant_id	All
REQ-P001-SEC-002	User SHALL have PSP-level role (PLATFORM_ADMIN, PSP_ADMIN, or PSP_OPS)	All
REQ-P001-SEC-003	Support Agent users SHALL have read-only access when support_scope = true	PSP_OPS
REQ-P001-SEC-004	All order data SHALL be scoped to user's tenant_id	All

2.3 Data Scope

- **Tenant Isolation:** Orders filtered by JWT tenant_id
- **Brand Visibility:** All brands within tenant visible
- **Campaign Scope:** All campaigns across brands visible

3. UI Components

3.1 Component Inventory

Component ID	Type	Description
P001-C001	Page Header	Title, order counts by status

P001-C002	Status Tabs	New, Acknowledged, All filter tabs
P001-C003	Search Bar	Order ID, store number search
P001-C004	Filter Panel	Brand, campaign, date range filters
P001-C005	Order Table	Sortable data grid with order list
P001-C006	Bulk Actions	Select-all, bulk acknowledge button
P001-C007	Order Detail Panel	Side panel with full order information
P001-C008	Pagination	Page navigation with count
P001-C009	Status Badge	Color-coded order status indicator

3.2 Layout Specification

+-----+						
	Order Queue					
	New: 12 Acknowledged: 8 Shipped: 45 Delivered: 234					
+-----+						
	[Search orders...]		[Brand ▼] [Campaign ▼]			
	[New (12)] [Acknowledged (8)] [All Orders]					
	[] Select All		[Acknowledge Selected]			
+-----+						
	[] Order #	Brand	Store	Items	Status	Age
+-----+						
	[] ORD-1234	Acme	STR-001	5	New	2h
	[] ORD-1235	Acme	STR-002	3	New	1h
	[] ORD-1236	Beta Co	STR-015	8	Ack	4h
+-----+						
	Showing 1-25 of 299		[< Prev] Page 1 of 12		[Next >]	
+-----+						

3.3 Component Specifications

P001-C001: Page Header

Property	Value
Title	"Order Queue"
Status Counts	Real-time counts by status enum
Refresh	Auto-refresh every 60 seconds

P001-C005: Order Table

Column	Field	Width	Sortable	Default Sort
--------	-------	-------	----------	--------------

Checkbox	selection	40px	No	-
Order #	order_number	120px	Yes	-
Brand	brand.name	150px	Yes	-
Store	store.store_number	120px	Yes	-
Items	line_count	80px	Yes	-
Status	status	120px	Yes	-
Age	created_at	100px	Yes	DESC (default)

4. Data Requirements

4.1 Entity Mapping

Entity	Table	Purpose
StoreOrder	store_orders	Primary order record
OrderLine	order_lines	Line items in order
Store	stores	Store information
Campaign	campaigns	Campaign reference
Brand	brands	Brand reference
KitItem	kit_items	Item details

4.2 Data Query

```
SELECT
  so.id, so.order_number, so.status, so.created_at,
  so.psp_order_ref, so.order_type,
  s.store_number, s.name as store_name,
  c.name as campaign_name,
  b.name as brand_name,
  COUNT(ol.id) as line_count,
  SUM(ol.quantity) as total_quantity
FROM store_orders so
JOIN stores s ON so.store_id = s.id
JOIN campaigns c ON so.campaign_id = c.id
JOIN brands b ON c.brand_id = b.id
LEFT JOIN order_lines ol ON ol.order_id = so.id
WHERE so.tenant_id = :tenant_id
      AND so.deleted_at IS NULL
      AND so.status IN (:status_filter)
GROUP BY so.id, s.id, c.id, b.id
```

ORDER BY so.created_at DESC
LIMIT :page_size OFFSET :offset

4.3 Data Requirements Matrix

Requirement ID	Description	Validation
REQ-P001-DATA-001	Order list SHALL display orders scoped to tenant	tenant_id filter
REQ-P001-DATA-002	Order list SHALL exclude soft-deleted records	deleted_at IS NULL
REQ-P001-DATA-003	Order age SHALL be calculated from created_at	Real-time calculation
REQ-P001-DATA-004	Line count SHALL reflect active order lines only	COUNT with deleted_at filter
REQ-P001-DATA-005	Status counts SHALL refresh on tab change	Query per status

5. Business Rules & Validation

5.1 Order Status Rules

Requirement ID	Rule	Enforcement
REQ-P001-BR-001	Orders in GENERATED status SHALL appear in "New" tab	Status filter
REQ-P001-BR-002	Orders in ACKNOWLEDGED status SHALL appear in "Acknowledged" tab	Status filter
REQ-P001-BR-003	Only GENERATED orders SHALL be selectable for bulk acknowledge	UI disable
REQ-P001-BR-004	Order status progression SHALL follow: GENERATED → ACKNOWLEDGED → IN_PRODUCTION → SHIPPED	State machine
REQ-P001-BR-005	Acknowledge action SHALL set acknowledged_at timestamp	Backend logic

5.2 Validation Rules

Requirement ID	Field	Validation	Error Message
REQ-P001-VAL-001	Order selection	At least one order selected for bulk action	"Select at least one order"
REQ-P001-VAL-002	Status transition	Order must be in valid source status	"Order cannot be acknowledged from current status"
REQ-P001-VAL-003	PSP Reference	Optional, max 50 characters	"PSP reference too long"

5.3 Business Constraints

Requirement ID	Constraint	Rationale
REQ-P001-BC-001	Orders SHALL be displayed FIFO (oldest first) in New tab	Production priority
REQ-P001-BC-002	Bulk acknowledge SHALL process maximum 100 orders per request	Performance limit
REQ-P001-BC-003	Order detail panel SHALL load within 500ms	UX requirement

6. API Integration Points

6.1 API Endpoints

Endpoint	Method	Purpose	Request	Response
/api/v1/orders	GET	List orders	Query params	OrderDTO[]
/api/v1/orders/{id}	GET	Order detail	Path param	OrderDetailDTO
/api/v1/orders/{id}/acknowledge	POST	Acknowledge order	OrderAckRequest	OrderDTO
/api/v1/orders/bulk-acknowledge	POST	Bulk acknowledge	BulkAckRequest	BulkAckResponse
/api/v1/orders/counts	GET	Status counts	Query params	CountsDTO

6.2 Request/Response Schemas

GET /api/v1/orders

Request Parameters:

Parameter	Type	Required	Description
status	string[]	No	Filter by status(es)
brand_id	uuid	No	Filter by brand
campaign_id	uuid	No	Filter by campaign
search	string	No	Search order_number, store_number
page	integer	No	Page number (default: 1)
page_size	integer	No	Items per page (default: 25, max: 100)
sort_by	string	No	Sort field
sort_order	string	No	ASC or DESC

Response Schema:

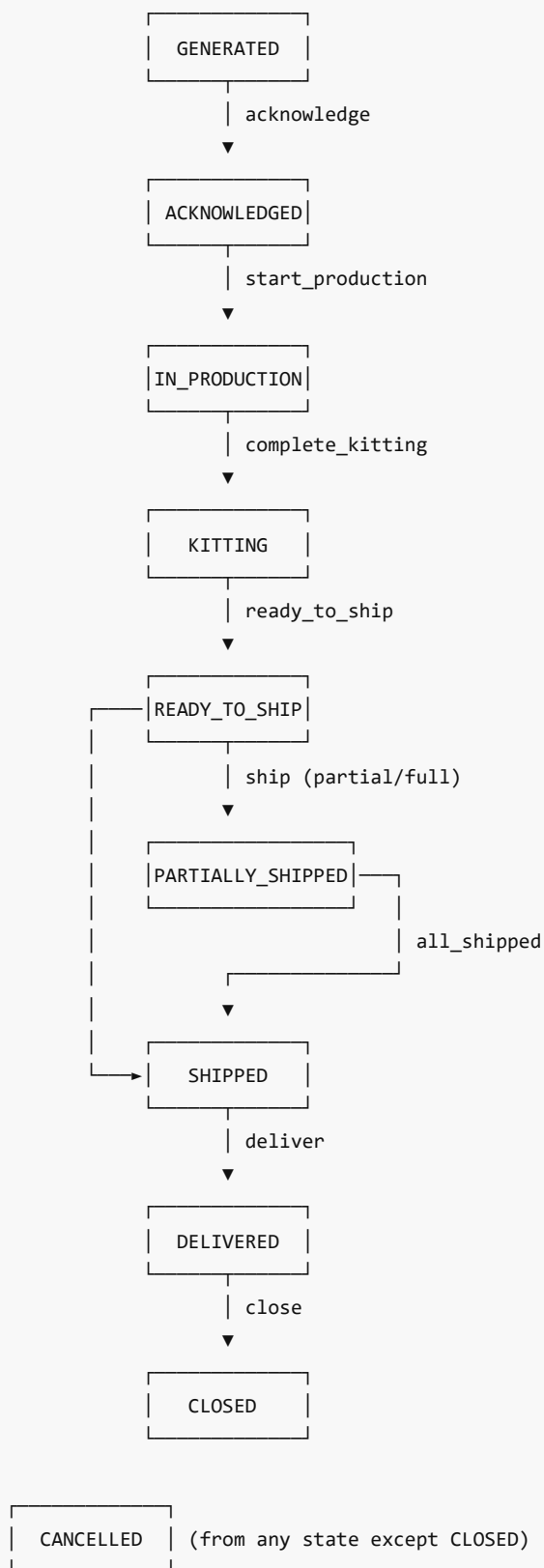
```
{
  "data": [
    {
      "id": "uuid",
      "order_number": "ORD-1234",
      "status": "GENERATED",
      "order_type": "INITIAL",
      "psp_order_ref": "PSP-REF-001",
      "created_at": "2025-12-15T10:30:00Z",
      "acknowledged_at": null,
      "store": {
        "id": "uuid",
        "store_number": "STR-001",
        "name": "Downtown Store"
      },
      "campaign": {
        "id": "uuid",
        "name": "Summer Promo"
      },
      "brand": {
        "id": "uuid",
        "name": "Acme Corp"
      },
      "line_count": 5,
      "total_quantity": 12
    }
  ],
  "pagination": {
    "page": 1,
    "page_size": 25,
    "total_items": 299,
    "total_pages": 12
  }
}
```

6.3 API Requirements

Requirement ID	Description	Implementation
REQ-P001-API-001	All API requests SHALL include Authorization header with JWT	Middleware
REQ-P001-API-002	Bulk acknowledge SHALL use Idempotency-Key header	Request header
REQ-P001-API-003	API responses SHALL include X-Request-ID for tracing	Response header
REQ-P001-API-004	Rate limiting SHALL apply: 100 requests/minute per user	API gateway

7. State Transitions

7.1 Order Status State Machine



7.2 State Transition Requirements

Requirement ID	Transition	From	To	Trigger
REQ-P001-ST-001	Acknowledge	GENERATED	ACKNOWLEDGED	User action
REQ-P001-ST-002	Start Production	ACKNOWLEDGED	IN_PRODUCTION	User action
REQ-P001-ST-003	Cancel	Any (not CLOSED)	CANCELLED	User action with reason
REQ-P001-ST-004	Ship	READY_TO_SHIP	SHIPPED/PARTIALLY_SHIPPED	Shipment creation
REQ-P001-ST-005	Deliver	SHIPPED	DELIVERED	Carrier webhook

7.3 Status Display Mapping

Status	Badge Color	Display Text
GENERATED	Yellow	New
ACKNOWLEDGED	Green	Acknowledged
IN_PRODUCTION	Blue	In Production
KITTING	Blue	Kitting
READY_TO_SHIP	Purple	Ready to Ship
PARTIALLY_SHIPPED	Orange	Partial Ship
SHIPPED	Blue	Shipped
DELIVERED	Gray	Delivered
CLOSED	Gray	Closed
CANCELLED	Red	Cancelled

8. Error Handling

8.1 Error Scenarios

Requirement ID	Error Scenario	HTTP Code	User Message	Recovery Action
REQ-P001-ERR-001	Unauthorized access	401	"Session expired. Please log in again."	Redirect to login
REQ-P001-ERR-002	Forbidden action	403	"You don't have permission to perform this action."	Display message

REQ-P001-ERR-003	Order not found	404	"Order not found."	Refresh list
REQ-P001-ERR-004	Invalid status transition	409	"Order cannot be acknowledged from current status."	Refresh order
REQ-P001-ERR-005	Bulk acknowledge partial failure	207	"X of Y orders acknowledged. Z failed."	Show failure details
REQ-P001-ERR-006	Server error	500	"Something went wrong. Please try again."	Retry with backoff
REQ-P001-ERR-007	Network timeout	-	"Connection lost. Retrying..."	Auto-retry 3x

8.2 Error Display

Component	Error Type	Display Method
Page Load	API failure	Full-page error with retry button
Table	Empty results	"No orders match your filters" message
Bulk Action	Partial failure	Toast notification with details link
Detail Panel	Load failure	Panel error state with retry

8.3 Logging Requirements

Requirement ID	Event	Log Level	Data
REQ-P001-LOG-001	Page load	INFO	user_id, filters
REQ-P001-LOG-002	Order acknowledge	INFO	user_id, order_id, prev_status
REQ-P001-LOG-003	Bulk acknowledge	INFO	user_id, order_ids, success_count
REQ-P001-LOG-004	Error	ERROR	error_code, message, stack_trace

9. Accessibility Requirements

9.1 WCAG 2.1 AA Compliance

Requirement ID	Guideline	Implementation
REQ-P001-A11Y-001	1.1.1 Non-text Content	All icons have aria-label
REQ-P001-A11Y-002	1.3.1 Info and Relationships	Table uses proper th/td semantics
REQ-P001-A11Y-003	1.4.1 Use of Color	Status uses color + text label
REQ-P001-A11Y-004	1.4.3 Contrast	4.5:1 minimum contrast ratio
REQ-P001-A11Y-005	2.1.1 Keyboard	All actions accessible via keyboard

REQ-P001-A11Y-006	2.4.1 Bypass Blocks	Skip navigation link provided
REQ-P001-A11Y-007	2.4.7 Focus Visible	Clear focus indicators
REQ-P001-A11Y-008	4.1.2 Name, Role, Value	ARIA roles for custom components

9.2 Keyboard Navigation

Key	Action
Tab	Navigate between interactive elements
Enter	Activate button/link, open detail panel
Space	Toggle checkbox selection
Escape	Close detail panel
Arrow Up/Down	Navigate table rows

9.3 Screen Reader Support

Component	Announcement
Status Tabs	"New orders tab, 12 items"
Order Row	"Order ORD-1234, Acme Corp, Store STR-001, 5 items, New, 2 hours ago"
Bulk Action	"Acknowledge 5 selected orders button"
Status Change	"Order ORD-1234 acknowledged" (live region)

10. Acceptance Criteria

10.1 Functional Acceptance Criteria

Criteria ID	Description	Verification Method
REQ-P001-AC-001	Order queue displays all orders for user's tenant	Query verification
REQ-P001-AC-002	Status tabs correctly filter orders by status	UI testing
REQ-P001-AC-003	Search filters orders by order number and store number	UI testing
REQ-P001-AC-004	Click on order row opens detail panel	UI testing
REQ-P001-AC-005	Acknowledge button updates order status to ACKNOWLEDGED	API + DB verification
REQ-P001-AC-006	Bulk acknowledge processes multiple orders	API + DB verification
REQ-P001-AC-007	Status counts update after acknowledge action	UI verification
REQ-P001-AC-008	Pagination correctly limits and navigates results	UI testing
REQ-P001-AC-009	Table sorting works for all sortable columns	UI testing

REQ-P001-AC-010	FIFO ordering maintained in New tab (oldest first)	Query verification
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10.2 Non-Functional Acceptance Criteria

Criteria ID	Description	Target	Verification
REQ-P001-AC-011	Page load time	< 2 seconds	Performance testing
REQ-P001-AC-012	Order detail panel load	< 500ms	Performance testing
REQ-P001-AC-013	Bulk acknowledge (100 orders)	< 5 seconds	Performance testing
REQ-P001-AC-014	WCAG 2.1 AA compliance	100%	Accessibility audit
REQ-P001-AC-015	Browser support	Chrome, Firefox, Edge, Safari	Cross-browser testing

10.3 Traceability Matrix

Requirement	Source	Test Case
REQ-P001-FR-001	SUPP-016	TC-P001-001
REQ-P001-SEC-001	SUPP-003	TC-P001-SEC-001
REQ-P001-DATA-001	3.1 Database Model	TC-P001-DATA-001
REQ-P001-API-001	3.4 Integration Architecture	TC-P001-API-001

Appendix A: Revision History

Version	Date	Author	Changes
1.0	2026-01-01	System	Initial specification

Document Status: Complete IEEE 830 Compliance: Section 3.2 - Functional Requirements

P002 - Shipments Screen Specification

Module: PSP Operations Portal **Screen ID:** P002 **Route:** /psp/shipments **Version:** 1.0 **Last Updated:** 2026-01-01 **IEEE 830 Reference:** Section 3.2 - Functional Requirements **SUPP Reference:** SUPP-016 (Fulfillment - Shipment Processing)

1. Screen Overview

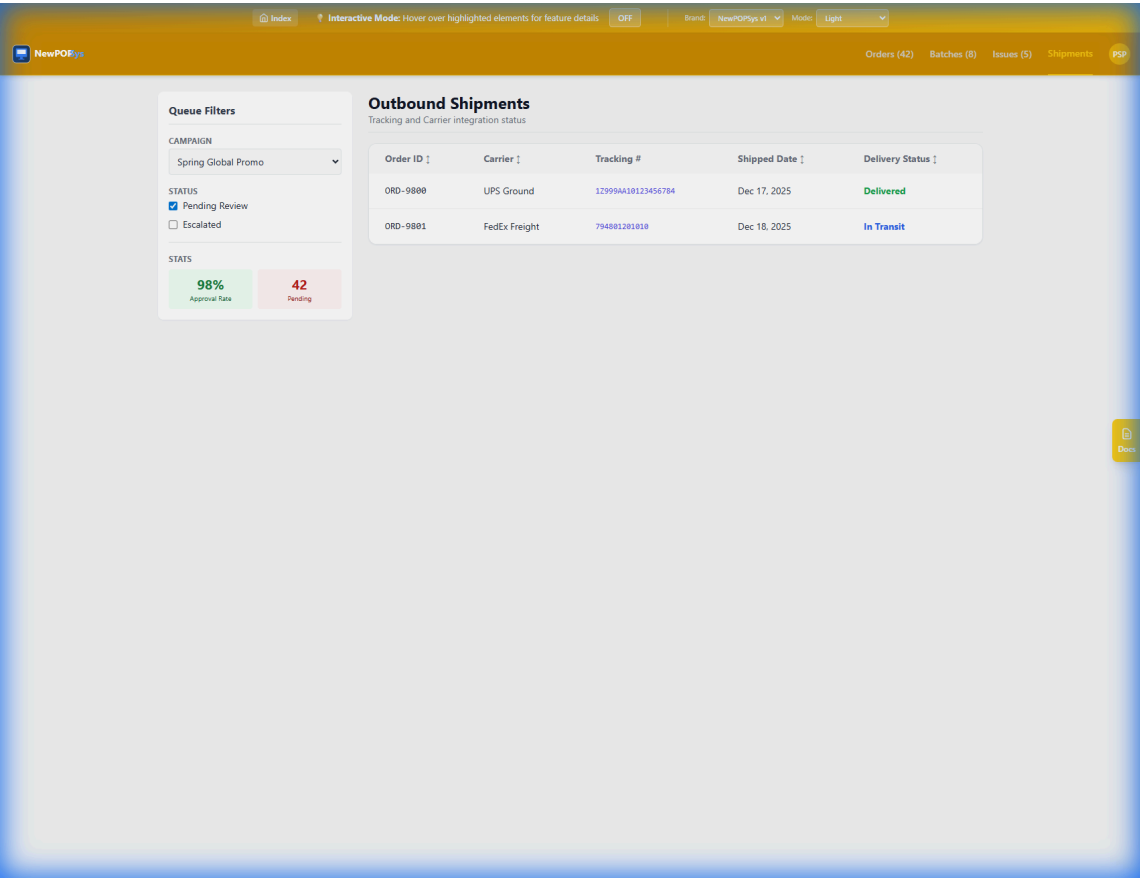
1.1 Purpose

The Shipments screen enables PSP Production Operators to create, track, and manage shipments for fulfilled orders. It provides carrier integration, tracking number management, and delivery status monitoring across all active shipments.

1.2 Business Context

This screen represents the final fulfillment stage where completed orders are shipped to stores. Shipment creation triggers webhook events to external systems and updates order status automatically. Carrier tracking integration provides real-time delivery visibility.

1.3 Screenshot Reference



1.4 Navigation Path

PSP Portal → Shipments (sidebar) → /psp/shipments

1.5 Related Screens

Screen	Relationship
P001 Order Queue	Orders ready for shipment

2. User Roles & Permissions

2.1 Authorized Roles

Role	Access Level	Permissions
PLATFORM_ADMIN	Full	View all, create/update shipments, void labels
PSP_ADMIN	Full	View all, create/update shipments, void labels
PSP_OPS	Operational	View all, create/update shipments
Support Agent (PSP_OPS + support_scope)	Read-Only	View only

2.2 Permission Requirements

Requirement ID	Description	Roles
REQ-P002-SEC-001	User SHALL be authenticated with valid JWT containing tenant_id	All
REQ-P002-SEC-002	User SHALL have PSP-level role for shipment operations	All
REQ-P002-SEC-003	Void label capability SHALL require PSP_ADMIN or higher	PSP_ADMIN, PLATFORM_ADMIN
REQ-P002-SEC-004	Support Agent SHALL have read-only access	PSP_OPS (support_scope)

2.3 Data Scope

- **Tenant Isolation:** Shipments filtered by JWT tenant_id
- **Order Linkage:** Only shipments for tenant's orders visible
- **Carrier Access:** Carrier credentials scoped to tenant

3. UI Components

3.1 Component Inventory

Component ID	Type	Description
P002-C001	Page Header	Title, shipment counts
P002-C002	Status Tabs	In Transit, Delivered, Exception, All
P002-C003	Search Bar	Tracking number, order number search
P002-C004	Filter Panel	Carrier, date range, status filters
P002-C005	Shipments Table	Sortable shipment list

P002-C006	Create Shipment Button	Opens shipment creation modal
P002-C007	Shipment Detail Panel	Side panel with tracking details
P002-C008	Create Shipment Modal	Form for new shipment entry
P002-C009	Tracking Link	External carrier tracking link
P002-C010	Carrier Logo	Visual carrier identification
P002-C011	Status Badge	Color-coded shipment status

3.2 Layout Specification

```
+-----+
| Shipments                                     [+ Create Shipment] |
| In Transit: 23 | Delivered: 156 | Exception: 2                |
+-----+
| [Search tracking #, order #...]   [Carrier ▼] [Date Range ▼] |
|                                   |                             |
| [In Transit (23)] [Delivered] [Exception (2)] [All]           |
|                                   |                             |
| +-----+ |
| | Tracking #   | Carrier | Order   | Store   | Status | ETA | |
| +-----+ |
| | 1Z999AA10... | [UPS]   | ORD-1234 | STR-001 | Transit | 12/18 | |
| | 7489012345... | [FEDEX] | ORD-1235 | STR-002 | Transit | 12/17 | |
| | 9400111899... | [USPS]  | ORD-1236 | STR-015 | Deliver | 12/15 | |
| | 1Z999AA10... | [UPS]   | ORD-1237 | STR-023 | Except  | -    | |
| +-----+ |
|                                   |                             |
| Showing 1-25 of 181                [< Prev] Page 1 of 8 [Next >] |
+-----+
```

3.3 Component Specifications

P002-C005: Shipments Table

Column	Field	Width	Sortable	Default Sort
Tracking #	tracking_numbers[0]	150px	Yes	-
Carrier	carrier	100px	Yes	-
Order	order.order_number	120px	Yes	-
Store	store.store_number	100px	Yes	-
Status	status	100px	Yes	-
ETA	estimated_delivery	100px	Yes	-
Created	created_at	120px	Yes	DESC (default)

P002-C008: Create Shipment Modal

+-----+
| Create Shipment [X] |
+-----+
|
| Order *
| [Search order number... ▼]
|
| Carrier *
| () UPS () FedEx () USPS () DHL
|
| Tracking Number(s) *
| +-----+
| | 1Z999AA10123456784 |
| +-----+
| [+ Add Another Tracking Number]
|
| Ship Date
| [2025-12-15]
|
| Items to Ship
| +-----+
	Item	Ordered	Shipping	
	Window Poster	2	[2]	
	Counter Disp	1	[1]	
	Shelf Talker	5	[5]	
+-----+				
[Cancel] [Create Shipment]				
+-----+

4. Data Requirements

4.1 Entity Mapping

Entity	Table	Purpose
Shipment	shipments	Shipment record
ShipmentLine	shipment_lines	Items in shipment
StoreOrder	store_orders	Parent order
OrderLine	order_lines	Order line items
Store	stores	Destination store
Campaign	campaigns	Campaign reference

4.2 Data Query


```

SELECT
  sh.id, sh.carrier, sh.tracking_numbers, sh.status,
  sh.shipped_at, sh.estimated_delivery, sh.delivered_at,
  so.order_number,
  s.store_number, s.name as store_name,
  c.name as campaign_name,
  COUNT(sl.id) as line_count
FROM shipments sh
JOIN store_orders so ON sh.order_id = so.id
JOIN stores s ON so.store_id = s.id
JOIN campaigns c ON so.campaign_id = c.id
LEFT JOIN shipment_lines sl ON sl.shipment_id = sh.id
WHERE sh.tenant_id = :tenant_id
  AND sh.deleted_at IS NULL
  AND sh.status IN (:status_filter)
GROUP BY sh.id, so.id, s.id, c.id
ORDER BY sh.created_at DESC
LIMIT :page_size OFFSET :offset

```

4.3 Data Requirements Matrix

Requirement ID	Description	Validation
REQ-P002-DATA-001	Shipment list SHALL be scoped to tenant	tenant_id filter
REQ-P002-DATA-002	Tracking numbers SHALL be stored as JSON array	JSONB column
REQ-P002-DATA-003	Carrier status SHALL sync from carrier API	Webhook/polling
REQ-P002-DATA-004	Partial shipments SHALL track quantity_shipped per line	shipment_lines.quantity_shipped
REQ-P002-DATA-005	ETA SHALL update from carrier tracking data	Carrier webhook

5. Business Rules & Validation

5.1 Shipment Creation Rules

Requirement ID	Rule	Enforcement
REQ-P002-BR-001	Shipment SHALL only be created for orders in READY_TO_SHIP or later status	API validation
REQ-P002-BR-002	Tracking number SHALL be unique across all shipments	Database constraint
REQ-P002-BR-003	Quantity shipped SHALL NOT exceed quantity ordered minus already shipped	API validation

REQ-P002-BR-004	At least one tracking number SHALL be provided	Form validation
REQ-P002-BR-005	Carrier SHALL be one of: UPS, FEDEX, USPS, DHL, OTHER	Enum validation

5.2 Validation Rules

Requirement ID	Field	Validation	Error Message
REQ-P002-VAL-001	order_id	Required, valid order	"Order is required"
REQ-P002-VAL-002	carrier	Required, valid enum	"Carrier is required"
REQ-P002-VAL-003	tracking_numbers	Required, array with 1+ items	"At least one tracking number required"
REQ-P002-VAL-004	tracking_numbers[]	Format validation per carrier	"Invalid tracking number format for carrier"
REQ-P002-VAL-005	quantity_shipped	Positive integer, <= remaining	"Quantity exceeds available"

5.3 Carrier Tracking Format Validation

Carrier	Format	Example
UPS	1Z + 16 alphanumeric	1Z999AA10123456784
FedEx	12-22 digits	748901234567
USPS	20-22 digits	9400111899223456789012
DHL	10 digits	1234567890

5.4 Business Constraints

Requirement ID	Constraint	Rationale
REQ-P002-BC-001	Shipment creation SHALL trigger webhook event	External system sync
REQ-P002-BC-002	Order status SHALL update to SHIPPED when all items shipped	Automatic transition
REQ-P002-BC-003	Order status SHALL update to PARTIALLY_SHIPPED if partial	Automatic transition
REQ-P002-BC-004	Delivered shipments SHALL NOT be editable	Data integrity

6. API Integration Points

6.1 API Endpoints

Endpoint	Method	Purpose	Request	Response
/api/v1/shipments	GET	List shipments	Query params	ShipmentDTO[]
/api/v1/shipments	POST	Create shipment	CreateShipmentRequest	ShipmentDTO
/api/v1/shipments/{id}	GET	Shipment detail	Path param	ShipmentDetailDTO
/api/v1/shipments/{id}	PATCH	Update shipment	UpdateShipmentRequest	ShipmentDTO
/api/v1/shipments/{id}/void	POST	Void shipment	-	VoidResponse
/api/v1/shipments/{id}/tracking	GET	Tracking events	Path param	TrackingEventDTO[]

6.2 Request/Response Schemas

POST /api/v1/shipments

Request Schema:

```
{
  "order_id": "uuid",
  "carrier": "UPS",
  "tracking_numbers": ["1Z999AA10123456784"],
  "shipped_at": "2025-12-15T10:00:00Z",
  "lines": [
    {
      "order_line_id": "uuid",
      "quantity_shipped": 2
    }
  ]
}
```

Response Schema:

```
{
  "id": "uuid",
  "carrier": "UPS",
  "tracking_numbers": ["1Z999AA10123456784"],
  "status": "LABEL_CREATED",
  "shipped_at": "2025-12-15T10:00:00Z",
  "estimated_delivery": null,
  "order": {
    "id": "uuid",
    "order_number": "ORD-1234"
  },
  "store": {
```

```
    "id": "uuid",
    "store_number": "STR-001"
  },
  "lines": [
    {
      "id": "uuid",
      "order_line_id": "uuid",
      "quantity_shipped": 2,
      "item_name": "Window Poster"
    }
  ],
  "created_at": "2025-12-15T10:00:00Z"
}
```

6.3 Webhook Events

Event	Trigger	Payload
shipment.created	New shipment created	ShipmentDTO
shipment.updated	Status/tracking updated	ShipmentDTO
shipment.delivered	Carrier confirms delivery	ShipmentDTO
shipment.exception	Delivery exception	ShipmentDTO + exception_reason

6.4 Carrier Integration

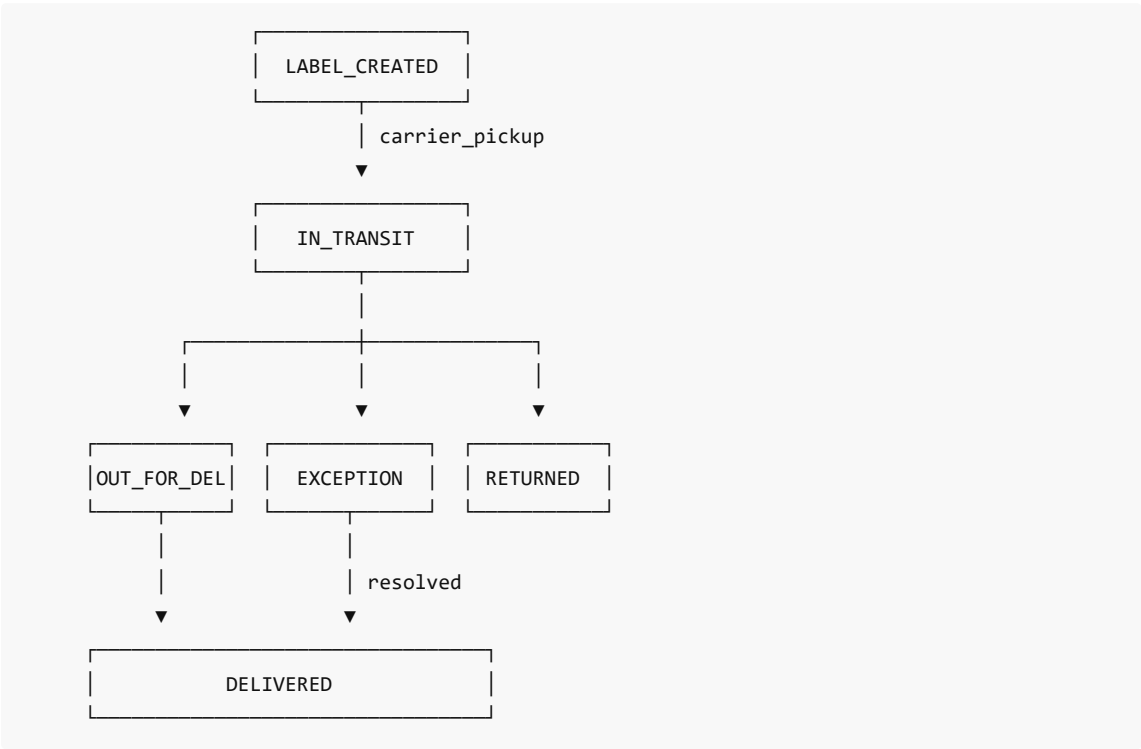
Requirement ID	Description	Implementation
REQ-P002-API-001	System SHALL support UPS, FedEx, USPS, DHL tracking	Carrier API adapters
REQ-P002-API-002	Tracking status SHALL poll every 4 hours	Background job
REQ-P002-API-003	Carrier webhooks SHALL update status in real-time when available	Webhook receiver
REQ-P002-API-004	Tracking link SHALL open carrier's tracking page	URL template per carrier

6.5 API Requirements

Requirement ID	Description	Implementation
REQ-P002-API-005	All write operations SHALL use Idempotency-Key	Request header
REQ-P002-API-006	Shipment creation SHALL emit webhook within 5 seconds	Async queue
REQ-P002-API-007	Rate limiting SHALL apply: 50 requests/minute per user	API gateway

7. State Transitions

7.1 Shipment Status State Machine



7.2 State Transition Requirements

Requirement ID	Transition	From	To	Trigger
REQ-P002-ST-001	Pickup	LABEL_CREATED	IN_TRANSIT	Carrier scan
REQ-P002-ST-002	Out for Delivery	IN_TRANSIT	OUT_FOR_DELIVERY	Carrier scan
REQ-P002-ST-003	Deliver	IN_TRANSIT/OUT_FOR_DELIVERY	DELIVERED	Carrier confirmation
REQ-P002-ST-004	Exception	IN_TRANSIT	EXCEPTION	Carrier exception event
REQ-P002-ST-005	Return	IN_TRANSIT	RETURNED	Return to sender event
REQ-P002-ST-006	Resolve	EXCEPTION	DELIVERED	Resolution + delivery

7.3 Status Display Mapping

Status	Badge Color	Display Text	Icon
LABEL_CREATED	Gray	Label Created	Tag
IN_TRANSIT	Blue	In Transit	Truck
OUT_FOR_DELIVERY	Purple	Out for Delivery	MapPin
DELIVERED	Green	Delivered	CheckCircle
EXCEPTION	Red	Exception	AlertTriangle
RETURNED	Orange	Returned	RotateCcw

8. Error Handling

8.1 Error Scenarios

Requirement ID	Error Scenario	HTTP Code	User Message	Recovery Action
REQ-P002-ERR-001	Invalid order for shipment	400	"Order is not ready for shipment."	Show order status
REQ-P002-ERR-002	Duplicate tracking number	409	"Tracking number already exists."	Clear field
REQ-P002-ERR-003	Quantity exceeds available	400	"Quantity exceeds remaining items."	Show max available
REQ-P002-ERR-004	Invalid tracking format	400	"Invalid tracking number format for carrier."	Show format hint
REQ-P002-ERR-005	Carrier API unavailable	503	"Carrier tracking temporarily unavailable."	Retry later
REQ-P002-ERR-006	Void failed	400	"Cannot void shipment in current status."	Show status
REQ-P002-ERR-007	Order already fully shipped	400	"All items already shipped."	Show shipments

8.2 Error Display

Component	Error Type	Display Method
Create Modal	Validation errors	Inline field errors
Create Modal	API errors	Banner at top of modal
Table	Load failure	Empty state with retry
Tracking	Carrier error	"Tracking unavailable" badge

8.3 Logging Requirements

Requirement ID	Event	Log Level	Data
REQ-P002-LOG-001	Shipment created	INFO	user_id, shipment_id, order_id
REQ-P002-LOG-002	Tracking update	INFO	shipment_id, old_status, new_status
REQ-P002-LOG-003	Carrier API call	DEBUG	carrier, tracking_number, response
REQ-P002-LOG-004	Webhook sent	INFO	shipment_id, webhook_id, event_type
REQ-P002-LOG-005	Error	ERROR	error_code, message, context

9. Accessibility Requirements

9.1 WCAG 2.1 AA Compliance

Requirement ID	Guideline	Implementation
REQ-P002-A11Y-001	1.1.1 Non-text Content	Carrier logos have alt text
REQ-P002-A11Y-002	1.3.1 Info and Relationships	Modal uses role="dialog" with aria-labelledby
REQ-P002-A11Y-003	1.4.1 Use of Color	Status uses color + icon + text
REQ-P002-A11Y-004	2.1.1 Keyboard	Modal traps focus when open
REQ-P002-A11Y-005	2.4.3 Focus Order	Logical tab order in modal
REQ-P002-A11Y-006	3.2.2 On Input	Form changes don't auto-submit
REQ-P002-A11Y-007	3.3.1 Error Identification	Errors associated with fields via aria-describedby
REQ-P002-A11Y-008	4.1.2 Name, Role, Value	Radio buttons properly grouped

9.2 Keyboard Navigation

Key	Action
Tab	Navigate between interactive elements
Enter	Activate button, submit modal
Space	Select radio button
Escape	Close modal, close detail panel
Arrow Keys	Navigate radio button group

9.3 Screen Reader Support

Component	Announcement
Create Modal	"Create Shipment dialog"

Carrier Radio	"UPS, radio button, 1 of 4"
Tracking Input	"Tracking number, required, text input"
Status Badge	"Status: In Transit"
Tracking Link	"Track shipment on UPS, opens in new tab"

10. Acceptance Criteria

10.1 Functional Acceptance Criteria

Criteria ID	Description	Verification Method
REQ-P002-AC-001	Shipment list displays all shipments for tenant	Query verification
REQ-P002-AC-002	Status tabs filter by shipment status	UI testing
REQ-P002-AC-003	Create Shipment button opens modal	UI testing
REQ-P002-AC-004	Modal validates required fields before submission	UI testing
REQ-P002-AC-005	Valid shipment creation saves to database	API + DB verification
REQ-P002-AC-006	Shipment creation triggers webhook event	Webhook log verification
REQ-P002-AC-007	Order status updates after shipment creation	DB verification
REQ-P002-AC-008	Tracking link opens carrier tracking page	UI testing
REQ-P002-AC-009	Partial shipment correctly updates remaining quantities	DB verification
REQ-P002-AC-010	Duplicate tracking number is rejected	API testing

10.2 Non-Functional Acceptance Criteria

Criteria ID	Description	Target	Verification
REQ-P002-AC-011	Page load time	< 2 seconds	Performance testing
REQ-P002-AC-012	Shipment creation	< 3 seconds	Performance testing
REQ-P002-AC-013	Webhook emission	< 5 seconds after create	Log verification
REQ-P002-AC-014	Carrier tracking sync	< 4 hour staleness	Monitoring
REQ-P002-AC-015	WCAG 2.1 AA compliance	100%	Accessibility audit

10.3 Integration Acceptance Criteria

Criteria ID	Description	Verification Method
REQ-P002-AC-016	UPS tracking numbers validated correctly	Unit testing
REQ-P002-AC-017	FedEx tracking numbers validated correctly	Unit testing

REQ-P002-AC-018	USPS tracking numbers validated correctly	Unit testing
REQ-P002-AC-019	Carrier status updates received via webhook	Integration testing
REQ-P002-AC-020	Delivery confirmation updates order status	Integration testing

10.4 Traceability Matrix

Requirement	Source	Test Case
REQ-P002-BR-001	SUPP-016	TC-P002-001
REQ-P002-SEC-001	SUPP-003	TC-P002-SEC-001
REQ-P002-DATA-001	3.1 Database Model	TC-P002-DATA-001
REQ-P002-API-001	3.4 Integration Architecture	TC-P002-API-001

Appendix A: Revision History

Version	Date	Author	Changes
1.0	2026-01-01	System	Initial specification

Document Status: Complete IEEE 830 Compliance: Section 3.2 - Functional Requirements

P003 - Issues Queue Screen Specification

Module: PSP Operations Portal **Screen ID:** P003 **Route:** /psp/issues **Version:** 1.0 **Last Updated:** 2026-01-01
IEEE 830 Reference: Section 3.2 - Functional Requirements **SUPP Reference:** SUPP-019 (Exception Management)

1. Screen Overview

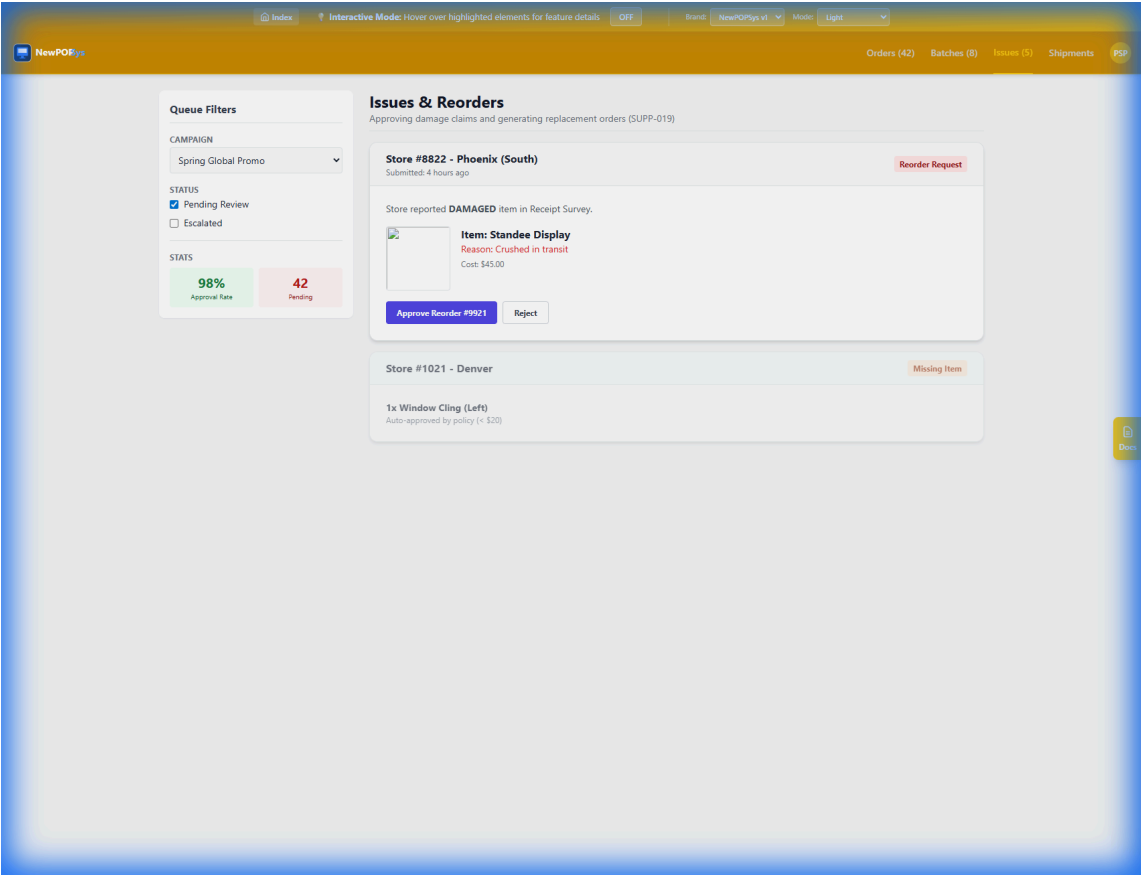
1.1 Purpose

The Issues Queue screen provides PSP Production Operators with a centralized view of all reported issues requiring triage and resolution. It enables issue review, approval/rejection decisions, and reorder creation for valid claims.

1.2 Business Context

This screen manages the exception handling workflow for store-reported problems including missing items, damaged goods, wrong items, and quantity shortages. Approved issues flow into reorder creation and fulfillment, closing the loop on store satisfaction.

1.3 Screenshot Reference



1.4 Navigation Path

PSP Portal → Issues (sidebar) → /psp/issues

1.5 Related Screens

Screen	Relationship
P001 Order Queue	Reorders appear in order queue
P002 Shipments	Reorder shipments tracked here
M03 Receipt Survey (Mobile)	Store reports issues from this screen

2. User Roles & Permissions

2.1 Authorized Roles

Role	Access Level	Permissions
PLATFORM_ADMIN	Full	View, triage, approve/reject, create reorders
PSP_ADMIN	Full	View, triage, approve/reject, create reorders

PSP_OPS	Operational	View, triage, approve/reject (scoped), create reorders
Support Agent (PSP_OPS + support_scope)	Read-Only	View only

2.2 Permission Requirements

Requirement ID	Description	Roles
REQ-P003-SEC-001	User SHALL be authenticated with valid JWT containing tenant_id	All
REQ-P003-SEC-002	User SHALL have PSP-level role for issue management	All
REQ-P003-SEC-003	PSP_OPS MAY approve issues under configured threshold	PSP_OPS
REQ-P003-SEC-004	PSP_ADMIN required for high-value issue approval	PSP_ADMIN, PLATFORM_ADMIN
REQ-P003-SEC-005	Rejection SHALL require reason and explanation	All with approve rights

2.3 Data Scope

- **Tenant Isolation:** Issues filtered by JWT tenant_id
- **Brand Visibility:** All brands within tenant visible
- **Store Access:** Issues from all stores visible

3. UI Components

3.1 Component Inventory

Component ID	Type	Description
P003-C001	Page Header	Title, issue counts by status
P003-C002	Status Tabs	Open, Triaged, In Fulfillment, Resolved
P003-C003	Search Bar	Issue ID, store number search
P003-C004	Filter Panel	Type, campaign, date range filters
P003-C005	Issues Table	Sortable issue list
P003-C006	Issue Detail Panel	Side panel with full issue information
P003-C007	Evidence Gallery	Photo viewer for issue evidence
P003-C008	Triage Actions	Approve, Reject, Request Info buttons
P003-C009	Reject Modal	Rejection reason form

P003-C010	Reorder Button	Create replacement order
P003-C011	Status Badge	Color-coded issue status
P003-C012	Type Badge	Issue type indicator

3.2 Layout Specification

```
+-----+
| Issues Queue                                     |
| Open: 8 | Triaged: 3 | In Fulfillment: 5 | Resolved: 127 |
+-----+
| [Search issues...] [Type ▼] [Campaign ▼]      |
|                                                 |
| [Open (8)] [Triaged (3)] [In Fulfillment (5)] [Resolved] |
|                                                 |
| +-----+ |
| | Issue # | Type   | Store  | Item      | Status | Age | |
| +-----+ |
| | ISS-1042 | DAMAGED | STR-001 | Window Post | Open   | 2h | |
| | ISS-1041 | MISSING | STR-015 | End Cap    | Open   | 4h | |
| | ISS-1040 | DAMAGED | STR-023 | Counter Dsp | Triaged | 1d | |
| | ISS-1038 | MISSING | STR-089 | Window Post | Fulfill | 2d | |
| | ISS-1035 | QTY_SHORT | STR-045 | Shelf Talk | Fulfill | 3d | |
| +-----+ |
|                                                 |
| Showing 1-25 of 143 [ < Prev ] Page 1 of 6 [ Next > ] |
+-----+
```

3.3 Issue Detail Panel

```
+-----+
| Issue ISS-1042 [X] |
+-----+
| Type: DAMAGED      |
| Status: OPEN       |
| Reported: Dec 15, 2025 at 10:30 AM |
| Age: 2 hours       |
|                   |
| Store: STR-001 - Acme Downtown |
| Campaign: Summer Promo |
|                   |
| Item Details       |
| -----           |
| Window Poster (24x36) |
| Qty Ordered: 2     |
| Qty Affected: 1    |
|                   |
| Description         |
| -----           |
| "Poster arrived with large tear across |
```

```

| the middle. Cannot be used for display." |
|                                           |
| Evidence Photos (2)                     |
| -----                                |
| [Photo 1] [Photo 2]                     |
|                                           |
| Triage Notes                            |
| +-----+                               |
| | Enter notes here...                    | |
| +-----+                               |
|                                           |
| [Reject] [Request Info] [Approve]       |
|                                           |
| After Approval:                         |
| [Create Reorder (1 unit)]               |
+-----+

```

3.4 Component Specifications

P003-C005: Issues Table

Column	Field	Width	Sortable	Default Sort
Issue #	issue_number	100px	Yes	-
Type	issue_type	100px	Yes	-
Store	store.store_number	100px	Yes	-
Item	kit_item.name	150px	Yes	-
Campaign	campaign.name	150px	Yes	-
Status	status	100px	Yes	-
Age	created_at	80px	Yes	ASC (FIFO)

4. Data Requirements

4.1 Entity Mapping

Entity	Table	Purpose
IssueRequest	issue_requests	Issue record
IssueLine	issue_lines	Items in issue (if multiple)
AssignmentItem	assignment_items	Original assignment
KitItem	kit_items	Item details
StoreAssignment	store_assignments	Store/campaign context
Store	stores	Store information

Campaign	campaigns	Campaign reference
PhotoUpload	photo_uploads	Evidence photos
ReorderRequest	reorder_requests	Linked reorder (if created)

4.2 Data Query

```
SELECT
  ir.id, ir.issue_number, ir.issue_type, ir.status,
  ir.description, ir.quantity_affected, ir.created_at,
  ir.triage_notes, ir.resolved_at, ir.resolution_type,
  ki.name as item_name, ki.sku,
  ai.quantity as qty_ordered,
  s.store_number, s.name as store_name,
  c.name as campaign_name,
  b.name as brand_name,
  (SELECT COUNT(*) FROM photo_uploads pu
   WHERE pu.reference_type = 'ISSUE'
   AND pu.reference_id = ir.id) as photo_count
FROM issue_requests ir
JOIN assignment_items ai ON ir.assignment_item_id = ai.id
JOIN kit_items ki ON ai.kit_item_id = ki.id
JOIN store_assignments sa ON ai.store_assignment_id = sa.id
JOIN stores s ON sa.store_id = s.id
JOIN campaigns c ON sa.campaign_id = c.id
JOIN brands b ON c.brand_id = b.id
WHERE ir.tenant_id = :tenant_id
      AND ir.deleted_at IS NULL
      AND ir.status IN (:status_filter)
ORDER BY ir.created_at ASC
LIMIT :page_size OFFSET :offset
```

4.3 Data Requirements Matrix

Requirement ID	Description	Validation
REQ-P003-DATA-001	Issue list SHALL be scoped to tenant	tenant_id filter
REQ-P003-DATA-002	Evidence photos SHALL be loaded on detail view	Lazy load
REQ-P003-DATA-003	Issue age SHALL be calculated from created_at	Real-time
REQ-P003-DATA-004	Related reorder SHALL be linked when created	FK reference
REQ-P003-DATA-005	FIFO ordering SHALL prioritize oldest issues	ORDER BY created_at ASC

5. Business Rules & Validation

5.1 Issue Type Rules

Requirement ID	Issue Type	Auto-Approve Eligible	Requires Photo
REQ-P003-BR-001	MISSING	Yes (if tracking shows delivered)	No
REQ-P003-BR-002	DAMAGED	No (requires photo review)	Yes
REQ-P003-BR-003	WRONG_ITEM	No (requires photo review)	Yes
REQ-P003-BR-004	QUANTITY_SHORT	Yes (if < ordered qty)	No

5.2 Triage Rules

Requirement ID	Rule	Enforcement
REQ-P003-BR-005	DAMAGED issues SHALL require at least one evidence photo	API validation
REQ-P003-BR-006	Rejection SHALL require reason code and explanation	Form validation
REQ-P003-BR-007	Approved issues SHALL be eligible for reorder creation	Status check
REQ-P003-BR-008	Reorder creation SHALL transition issue to IN_FULFILLMENT	State machine
REQ-P003-BR-009	Reorder delivery SHALL auto-resolve issue	Webhook trigger

5.3 Validation Rules

Requirement ID	Field	Validation	Error Message
REQ-P003-VAL-001	triage_notes	Max 1000 characters	"Notes too long"
REQ-P003-VAL-002	rejection_reason	Required for reject	"Rejection reason required"
REQ-P003-VAL-003	rejection_explanation	Required for reject, 10+ chars	"Please provide explanation"
REQ-P003-VAL-004	reorder_quantity	1 to quantity_affected	"Invalid quantity"

5.4 Rejection Reason Codes

Code	Display	Description
INSUFFICIENT_EVIDENCE	Insufficient evidence	Photos don't show claimed issue
ITEM_USABLE	Item appears usable	Damage doesn't affect display
OUTSIDE_WINDOW	Outside return window	Reported too late
DUPLICATE	Duplicate request	Already submitted
OTHER	Other	Free-form explanation required

6. API Integration Points

6.1 API Endpoints

Endpoint	Method	Purpose	Request	Response
/api/v1/issues	GET	List issues	Query params	IssueDTO[]
/api/v1/issues/{id}	GET	Issue detail	Path param	IssueDetailDTO
/api/v1/issues/{id}/triage	POST	Triage decision	TriageRequest	IssueDTO
/api/v1/issues/{id}/approve	POST	Approve issue	ApproveRequest	IssueDTO
/api/v1/issues/{id}/reject	POST	Reject issue	RejectRequest	IssueDTO
/api/v1/issues/{id}/request-info	POST	Request more info	RequestInfoDTO	IssueDTO
/api/v1/issues/{id}/reorder	POST	Create reorder	ReorderRequest	ReorderDTO
/api/v1/issues/{id}/photos	GET	Evidence photos	Path param	PhotoDTO[]
/api/v1/issues/counts	GET	Status counts	Query params	CountsDTO

6.2 Request/Response Schemas

POST /api/v1/issues/{id}/reject

Request Schema:

```
{
  "reason_code": "INSUFFICIENT_EVIDENCE",
  "explanation": "Photos do not clearly show the reported damage. The item appears intact.",
  "notify_store": true
}
```

Response Schema:

```
{
  "id": "uuid",
  "issue_number": "ISS-1042",
  "status": "REJECTED",
  "issue_type": "DAMAGED",
  "rejection": {
    "reason_code": "INSUFFICIENT_EVIDENCE",
    "explanation": "Photos do not clearly show the reported damage.",
    "rejected_by": "user@psp.com",
    "rejected_at": "2025-12-15T14:30:00Z"
  },
  "store_notified": true
}
```

POST /api/v1/issues/{id}/reorder

Request Schema:

```
{
  "quantity": 1,
  "notes": "Replacement for damaged item",
  "priority": "NORMAL"
}
```

Response Schema:

```
{
  "id": "uuid",
  "reorder_number": "REO-1042",
  "status": "PENDING",
  "issue_id": "uuid",
  "order": {
    "id": "uuid",
    "order_number": "ORD-1500",
    "status": "GENERATED"
  },
  "quantity": 1,
  "created_at": "2025-12-15T14:35:00Z"
}
```

6.3 Webhook Events

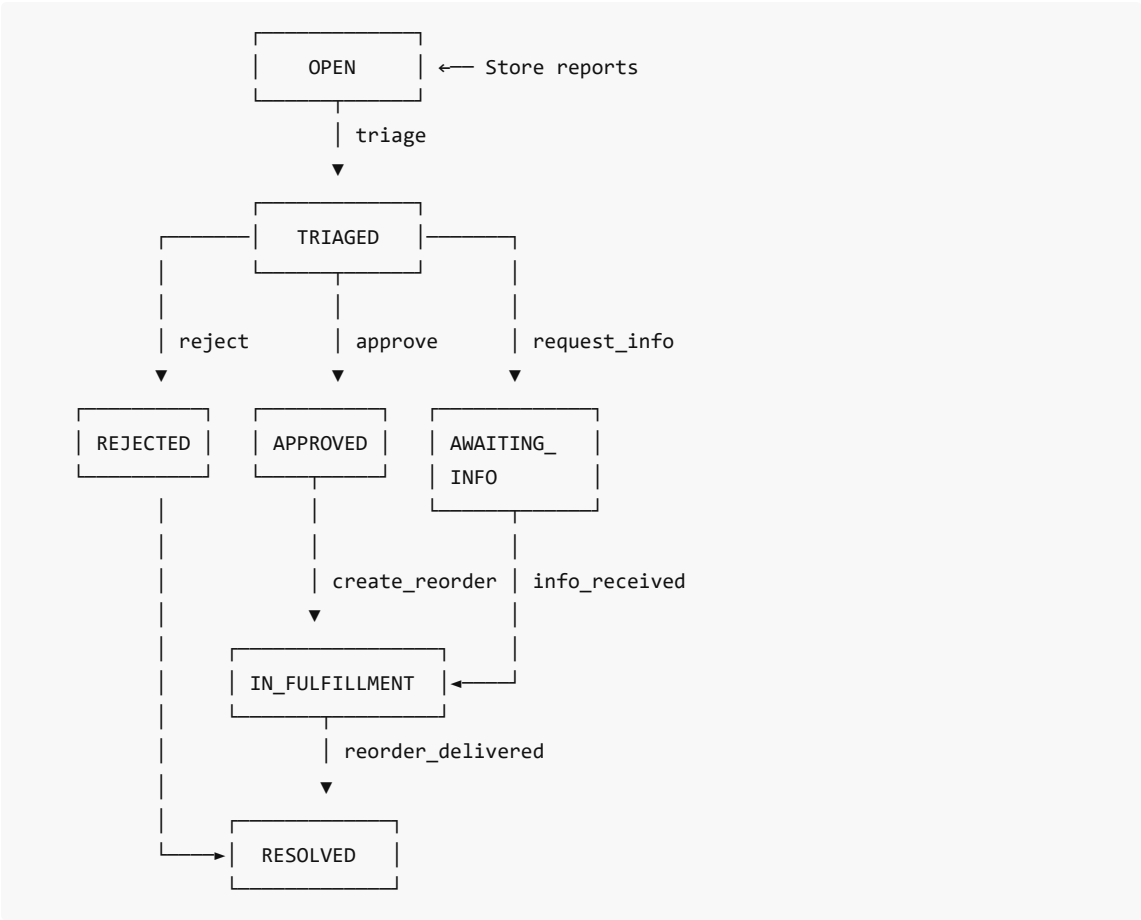
Event	Trigger	Payload
issue.submitted	Store submits issue	IssueDTO
issue.triaged	PSP triages issue	IssueDTO
issue.approved	PSP approves issue	IssueDTO
issue.rejected	PSP rejects issue	IssueDTO + rejection
issue.resolved	Reorder delivered	IssueDTO + resolution

6.4 API Requirements

Requirement ID	Description	Implementation
REQ-P003-API-001	All decisions SHALL emit webhook events	Async queue
REQ-P003-API-002	Reject SHALL notify store via email/push	Notification service
REQ-P003-API-003	Reorder creation SHALL generate new order	Order service
REQ-P003-API-004	Idempotency-Key required for all writes	Request header

7. State Transitions

7.1 Issue Status State Machine



7.2 State Transition Requirements

Requirement ID	Transition	From	To	Trigger
REQ-P003-ST-001	Triage	OPEN	TRIAGED	PSP reviews
REQ-P003-ST-002	Approve	TRIAGED	APPROVED	PSP approves
REQ-P003-ST-003	Reject	TRIAGED	REJECTED	PSP rejects
REQ-P003-ST-004	Request Info	TRIAGED	AWAITING_INFO	Need more details
REQ-P003-ST-005	Info Received	AWAITING_INFO	TRIAGED	Store responds
REQ-P003-ST-006	Create Reorder	APPROVED	IN_FULFILLMENT	Reorder created
REQ-P003-ST-007	Resolve	IN_FULFILLMENT	RESOLVED	Reorder delivered
REQ-P003-ST-008	Deny Resolution	REJECTED	RESOLVED	Rejection finalized

7.3 Status Display Mapping

Status	Badge Color	Display Text	Icon
--------	-------------	--------------	------

OPEN	Yellow	Open	AlertCircle
TRIAGED	Green	Triaged	Eye
APPROVED	Green	Approved	CheckCircle
AWAITING_INFO	Orange	Awaiting Info	HelpCircle
IN_FULFILLMENT	Blue	In Fulfillment	Package
RESOLVED	Gray	Resolved	Check
REJECTED	Red	Rejected	XCircle

7.4 Issue Type Display

Type	Badge Color	Icon	Description
MISSING	Red	XCircle	Item not received
DAMAGED	Orange	AlertTriangle	Item unusable
WRONG_ITEM	Purple	HelpCircle	Different item received
QUANTITY_SHORT	Yellow	Minus	Fewer than expected

8. Error Handling

8.1 Error Scenarios

Requirement ID	Error Scenario	HTTP Code	User Message	Recovery Action
REQ-P003-ERR-001	Issue not found	404	"Issue not found."	Refresh list
REQ-P003-ERR-002	Invalid status transition	409	"Cannot perform action on issue in current status."	Show current status
REQ-P003-ERR-003	Rejection without reason	400	"Please select a rejection reason."	Highlight field
REQ-P003-ERR-004	Reorder for unapproved issue	400	"Issue must be approved before creating reorder."	Show approve button
REQ-P003-ERR-005	Duplicate reorder	409	"Reorder already exists for this issue."	Show reorder link
REQ-P003-ERR-006	Photo load failed	500	"Unable to load evidence photos."	Retry button
REQ-P003-ERR-007	Permission denied	403	"You don't have permission to approve this issue."	Show requirements

8.2 Error Display

Component	Error Type	Display Method
Detail Panel	Load error	Panel error state
Triage Actions	Validation error	Button tooltip
Reject Modal	Form error	Inline field errors
Photo Gallery	Load error	Placeholder with retry

8.3 Logging Requirements

Requirement ID	Event	Log Level	Data
REQ-P003-LOG-001	Issue viewed	INFO	user_id, issue_id
REQ-P003-LOG-002	Issue approved	INFO	user_id, issue_id, notes
REQ-P003-LOG-003	Issue rejected	INFO	user_id, issue_id, reason_code
REQ-P003-LOG-004	Reorder created	INFO	user_id, issue_id, reorder_id
REQ-P003-LOG-005	Error	ERROR	error_code, message, context

9. Accessibility Requirements

9.1 WCAG 2.1 AA Compliance

Requirement ID	Guideline	Implementation
REQ-P003-A11Y-001	1.1.1 Non-text Content	Evidence photos have descriptive alt text
REQ-P003-A11Y-002	1.3.1 Info and Relationships	Panel sections use proper heading hierarchy
REQ-P003-A11Y-003	1.4.1 Use of Color	Status and type use color + text + icon
REQ-P003-A11Y-004	2.1.1 Keyboard	All actions accessible via keyboard
REQ-P003-A11Y-005	2.4.4 Link Purpose	Action buttons have descriptive labels
REQ-P003-A11Y-006	3.2.2 On Input	Modal actions require explicit submit
REQ-P003-A11Y-007	3.3.2 Labels	All form fields have visible labels
REQ-P003-A11Y-008	4.1.3 Status Messages	Status changes announced to screen readers

9.2 Keyboard Navigation

Key	Action
Tab	Navigate between interactive elements
Enter	Activate button, open detail panel

Escape	Close panel, close modal
Arrow Left/Right	Navigate photo gallery
Space	Toggle photo full-screen

9.3 Screen Reader Support

Component	Announcement
Issue Row	"Issue ISS-1042, Damaged, Store STR-001, Window Poster, Open, 2 hours ago"
Status Badge	"Status: Open"
Type Badge	"Issue type: Damaged"
Photo Gallery	"Evidence photo 1 of 2"
Action Buttons	"Approve issue button", "Reject issue button"
Status Change	"Issue ISS-1042 approved" (live region)

10. Acceptance Criteria

10.1 Functional Acceptance Criteria

Criteria ID	Description	Verification Method
REQ-P003-AC-001	Issues queue displays all active issues for tenant	Query verification
REQ-P003-AC-002	Status tabs correctly filter issues by status	UI testing
REQ-P003-AC-003	FIFO ordering shows oldest issues first	Query verification
REQ-P003-AC-004	Click on issue row opens detail panel	UI testing
REQ-P003-AC-005	Evidence photos load in detail panel	UI testing
REQ-P003-AC-006	Approve button transitions issue to APPROVED	API + DB verification
REQ-P003-AC-007	Reject requires reason code and explanation	UI testing
REQ-P003-AC-008	Rejection sends notification to store	Notification verification
REQ-P003-AC-009	Create Reorder generates new order	API + DB verification
REQ-P003-AC-010	Reorder delivery auto-resolves issue	Integration testing

10.2 Business Logic Acceptance Criteria

Criteria ID	Description	Verification Method
REQ-P003-AC-011	DAMAGED issues require photo evidence	API validation testing
REQ-P003-AC-012	MISSING issues may auto-approve if tracking confirms	Business rule testing

REQ-P003-AC-013	Quantity in reorder does not exceed affected quantity	API validation testing
REQ-P003-AC-014	Issue status follows defined state machine	State transition testing
REQ-P003-AC-015	Resolved issues move to history tab	UI testing

10.3 Non-Functional Acceptance Criteria

Criteria ID	Description	Target	Verification
REQ-P003-AC-016	Page load time	< 2 seconds	Performance testing
REQ-P003-AC-017	Detail panel load	< 500ms	Performance testing
REQ-P003-AC-018	Photo gallery load	< 2 seconds	Performance testing
REQ-P003-AC-019	WCAG 2.1 AA compliance	100%	Accessibility audit
REQ-P003-AC-020	Browser support	Chrome, Firefox, Edge, Safari	Cross-browser testing

10.4 Traceability Matrix

Requirement	Source	Test Case
REQ-P003-BR-001	SUPP-019	TC-P003-001
REQ-P003-SEC-001	SUPP-003	TC-P003-SEC-001
REQ-P003-DATA-001	3.1 Database Model	TC-P003-DATA-001
REQ-P003-API-001	3.4 Integration Architecture	TC-P003-API-001
REQ-P003-ST-001	SUPP-019 State Machine	TC-P003-ST-001

Appendix A: Revision History

Version	Date	Author	Changes
1.0	2026-01-01	System	Initial specification

Document Status: Complete IEEE 830 Compliance: Section 3.2 - Functional Requirements